

Company: IDFC First Bank

Job Profile: Application Engineer

Offer Type: Internship + Placement

Internship Stipend: Not specified yet

CTC: 18.10 LPA (Fixed: 14 LPA + Joining Bonus: 2 L + Variable: 2.10 LPA)

Location: Mumbai/Bangalore/Hyderabad

Process Date: PPT - 25/07 (online)

OA - 28/07

Interviews - 30/07

Placement Process:

Online Assessment: this was the first round which consisted of 3 sections

1. general aptitude / reasoning - 20 min
2. Core cs fundamentals - 30 min
3. Coding Section - 40 min

Core cs fundamentals involved mcqs based on code snippets/flowcharts, DBMS, OS, SQL. It was completely language independent and basic level questions were asked.

Coding section involved two easy questions-

1. Given an array of integers, return the count of correctly placed and incorrectly placed integers if the array was sorted. It was quite easy and the attached problem is almost similar to it. ([link](#))
2. Given 2 integers that represent a range from x to y, find the sum of all the golden numbers where a golden number is defined as a number whose sum of digits - maximum digit is equal to maximum digit (eg- 352: $3 + 5 + 2 = 10$, max digit = 5, so $10 - 5 = 5$ ie sum-maxd = maxd). I was not able to pass all the test cases idk if there was some issue with setting the testcase (did 7/10). This was the crux but the question was in a story format like we encounter over codeforces(so better if you are already familiar with that type)

TIPS:

- Time management is very important in the first 2 sections. Questions are doable but the decision to attempt or skip a particular question based upon time is more important.
- Many python users faced issues in this round and this was seen in PhonePe as well, so if you've enough time consider shifting to cpp/java

1. **Technical Round 1** [37 candidates appeared for this round]

This round was entirely about the basics of OOP and core subjects (OS and CCN) and schema design. It started with a short introduction followed by OOP fundamentals in your first language, mine was cpp so in depth questions related to friend functions/virtual keyword/multiple inheritance/destructors were asked followed by some questions related to OS and CCN. I was also asked about the basics of Cloud computing even though I didn't have any project related to it. So in this case answer honestly whatever you know and try to relate concepts.

Later the interviewer asked me to design a schema for an e-commerce platform, followed by how would you design a recommendation system over this and how would you store overhead data specifically in which tables. I was then asked to write a basic SQL query over these tables using 2 joins.

This was extended into a discussion related to system design: how would you scale this, what databases will you use(sql/nosql/both with reasoning) and how would you prevent bruteforce attacks(IP blacklisting using Redis). This round went on for about 35-40 mins.

TIPS:

- Make sure you are clear with the basics of your first language and have some idea of OOP in Java too if java isn't your first language. Also never ignore OOP in your first language, even though you've mastered it in Java, have a basic understanding of how that works in your preferred language.
- Also from what I got to know, candidates who had their first language as Java, were also asked about development in Java (Springboot related) so be prepared to handle such situations.
- Design problems are to check how you approach a problem so take your time and think in 360 degree just to not miss out on any part. Make sure you ask genuine questions related to company tech at the end.

2. **Technical Round 2** [22 candidates appeared for this round]

This round was focused more towards resume and Low Level Design(LLD). After introduction I was drilled down on a single project where they even asked me how do you write config files, how do you build docker images and configure kubernetes and everything, so make sure you do projects yourself as they also check github in some cases. They also asked what all types of databases you've worked on apart from mongoDB and Mysql and what are their use cases(vector db and key value pair db like redis)

In the later part I was asked to give a low level design of a cab booking system (Ola/Uber). So in LLD, they expect you to use OOP and data structures and algorithms and I was asked to write code structure in cpp (classes, functions they will have, what each func will do and which data structure and algo it will use). Keep discussing with the interviewer what all things you are considering, like at the first step I asked if I had to design for both drivers and passengers so he said no continue only with passengers.

So I went on to do it that way and told them that Dijkstra's would be required but I won't simply apply it over all the drivers. I will set a radius range r initially small(lets say 500m) and if I found drivers within that range they would be sent to client side(passenger) and if not found the radius would be increased to say 1km and the process would go on until a driver is encountered or we exceed the max radius which can be set to 5km. Along with this, I would also use a priority queue to keep track of "k" nearest drives in case the first one rejects or something(they see how you handle edge cases). I also mentioned that we would need to

implement a lock in a way so that 2 passengers don't book the same cab for diff destinations and explained how that part would be implemented in the driver part of the design. Then they asked follow up questions like what if multiple passengers are requesting the same driver then which passenger would be allocated to the driver (which actually happens in the real world). We kept discussing and enhancing the system and this round went on for 30 min or so.

TIPS:

- Yes surprisingly most of us were never asked about DSA problems.
- **For all your projects make a google doc with all the important points/features, their implementation and tech stack** as it becomes really easy to revise the entire project cause at the last moment we feel we need to revise everything.
- Also add to it what problems did you faced and how did you resolve them as it is a check question if you've done the project yourself or not
- Whenever you feel confused, communicate with the interviewer, they'll surely help you out. It's okay even if you ask them if you're doing right or not so they can guide you well to produce a solution.

3. **HR Round** [9 candidates who appeared for this round]

It was a classic HR round, more of a conversation type where I was asked some basic questions like why do you want to join IDFC and all. I was given situations and was asked how would you resolve this. Here mainly they check if the candidate is culturally fit and would be able to coordinate with the team so just keep a good smile and try to connect answers to your own experiences. Mention about the ppt/research that you do which adds more weightage to your answer.

I was asked the following questions-

1. What one thing could piss you off at work?
2. How would you deal with a team mate who is not coordinating/communicating?
3. If you're a manager and supposed to give a best team member award, what criteria would you consider before deciding the winner apart from tech/development?

TIPS:

- Try to connect answers with company values. Read about their vision/mission, tagline, history, this really helps in HR based rounds.
- There might be some trick questions which play on our psychology so it is always safer to answer honestly.
- Be friendly with the interviewer and reflect your true personality.

4. **Online Screening Round** [9 candidates who appeared for this round]

It was an extra round conducted online which started at 8:30pm. It was just a fitment round where I was asked to explain my projects to a very senior positioned interviewer which lasted for just 15 mins. There

I was directly asked to send the repo link of my project and explain it. Keep in mind don't take such a round lightly as elimination does take place here which unfortunately did happen for one of my peers. Some candidates were asked about the subjects in previous semester or the course work, so make sure you have an idea of that as well, they don't expect details but just the list of subjects and topics that you've learned.

The results were declared the next day and 8 students received the offer.

Useful Tips:

1. Prepare basics really well, not being able to answer them gives a bad impression.
2. To be very honest, don't consider DSA as the only way to conquer placements. This year saw the rise in the difficulty level so try doing CP but make sure to set a balance between dev/dsa and start revising core subjects early.
3. Basic web dev projects won't be enough, it's all saturated and AI/vibe coding is escalating at a very high rate so try exploring basics of devops or app dev.
4. What I observed this year was that many companies were asking LLD/system design, so once you are done with all the preparation do consider giving time to these topics as well.
5. Give mock interviews with friends or even seniors as it helps build confidence and give us a clear idea of our weak sections.
6. Never let rejections bring you down as it may affect the upcoming opportunity. The process is tiring and requires a lot of patience. There's a lot of luck involved, so focus on what you can control and leave the rest to destiny.
7. **Don't repeat my mistakes**
 - Get an internship even if it's unpaid!!!
 - Consider learning LLD/system design early on, I only looked into it at the last moment and couldn't go in-depth.

Resources-

System Design - [\[Complete\] System Design Roadmap with Videos for SDEs - Interviews](#)

Low Level Design (LLD) - [Low Level Design | System Design | codeWithAryan](#)

Feel free to contact me: [Shrihari Joshi | LinkedIn](#)

Resume which I'd used:  [Shrihari_Joshi_019.pdf](#)

All the best!!!! You'll do well. Have faith in yourself!